ANTRA NAYUDU

MACHINE LEARNING ENGINEER | COMPUTER VISION

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PROFESSIONAL SUMMARY

Machine learning engineer specializing in research-oriented data workflows, from web scraping and entity resolution to modeling, evaluation, and production-ready data pipelines. Leverages Python, Pandas, NumPy, scikit-learn, PyTorch, NLP and prompt engineering, PostgreSQL, Docker, Git, MLflow, and strong technical documentation to build reproducible analytical and investigative datasets.

TECHNICAL SKILLS

Programming: Python, SQL, Web Scraping, HTTP And HTML Parsing
ML & Computer Vision: PyTorch, TensorFlow, scikit-learn, OpenCV, YOLOv8, EfficientNet, semantic segmentation, transfer learning
Data & ML Lifecycle: Pandas, NumPy, CVAT annotation, data cleaning, augmentation, feature engineering, model validation, hyperparameter tuning, PostgreSQL
Operations & Tools: MLflow experiment tracking, periodic retraining workflows, AWS, AWS S3, Git, Matplotlib, technical documentation, Docker
Generative AI / NLP: T5-small, DistilBERT, LSTM, transcript summarization and sentiment-classification experiments, Natural Language Processing (NLP), Prompt Engineering

EXPERIENCE

Oracle Lens — **Oct 2025 - Apr 2026**
AI/ML Researcher - Financial Forecasting
- Built a Python time-series forecasting workflow that ingested historical and live market data from the Alpaca Markets API, then applied data cleaning and feature engineering across lag, rolling-statistic, trend, and seasonality variables.
- Structured 30-60 timestep lookback windows and 1-7 day forecast horizons, using RMSE and MAPE to validate model performance and communicate analytical results.
- Managed experiment and model versions with MLflow while implementing a periodic retraining pipeline that incorporated newly ingested market data.

Kevares — **Oct 2024 - Apr 2025**
AI & ML Researcher (Co-op) - Computer Vision
- Engineered a PyTorch and YOLOv8 semantic-segmentation pipeline for curb detection, supporting automated analysis of road and built-environment imagery.
- Annotated and quality-checked 1,672 images in CVAT, diagnosed data-quality limitations, and strengthened the training dataset through a more robust annotation strategy and external data source.
- Benchmarked YOLOv8 nano, small, and medium configurations, manually tuning hyperparameters and documenting accuracy, efficiency, failure modes, and practical model tradeoffs through reports, visualizations, and stakeholder presentations.

Learn and Empower Pvt. Ltd. — **Jun 2023 - Jun 2024**
Machine Learning Intern - Computer Vision
- Developed a Python-based Indian Sign Language-to-text prototype with OpenCV, TensorFlow, and transfer learning, producing real-time text predictions for an accessibility use case.
- Collected, cleaned, transformed, and validated 13,974 images across 38 gesture classes, creating a consistent dataset for repeatable training and evaluation.
- Investigated misclassifications, refined image preprocessing and training inputs, iterated on an EfficientNet-based classifier, and documented methodology, evaluation results, and improvement decisions with cross-functional stakeholders.

SELECTED ML EXPERIMENTATION

- **Yelp Review Sentiment Analysis:** Compared BiLSTM and DistilBERT models for three-class sentiment classification, using validation performance and error analysis to select the stronger approach.
- **Minutes by Meeting:** Fine-tuned T5-small to convert meeting transcripts into concise summaries and action-focused bullet points, with sentiment analysis to capture the meeting’s overall tone.

EDUCATION

Humber Polytechnic — **2025 - 2026**
Graduate Certificate, Artificial Intelligence with Cybersecurity

Humber Polytechnic — **2024 - 2025**
Graduate Certificate, Artificial Intelligence & Machine Learning

G.H. Raisoni College of Engineering — **2020 - 2024**
Bachelor of Engineering, Computer Science
- **Achievements:** Dean's List of Honour